

Centre of Australian Category Theory (CoACT)
A Macquarie University Research Centre

Annual Report for 2004

I. List of members

Advisory Board

Professor Ross Street	Director of CoACT
Professor Michael Johnson	Associate Director of CoACT
Emeritus Professor Max Kelly	University of Sydney
Professor Ray Offen (<i>ex officio</i>)	Dean, Division of ICS, Macquarie University
Dr Wesley Phoa	Capital Strategy Research, USA
Associate Professor Dominic Verity	Director, Postgraduate Coursework Programs, Division of ICS, Macquarie University
Professor Di Yerbury (<i>ex officio</i>)	Vice-Chancellor, Macquarie University

Other Members

Dr Michael Batanin	Scott Russell Johnson Memorial Fellow, Math Dept
Dr John Corbett	Senior Research Fellow, Math Dept
Dr Alexei Davydov	Research Fellow, Math Dept
Dr Brian Day	Research Associate, Math Dept
Dr Lee Flax	Senior Lecturer, Computing Dept
Ms Carolyn Kennett	Director, Macquarie University Numeracy Centre
Dr Steve Lack	Lecturer, School of Quantitative Methods and Mathematical Sciences, University of Western Sydney
Ms Catherine Menon	PhD Student, Computing Dept
Ms Margaret Mitchell	PhD Student, Computing Dept
Mr Elango Panchadcharam	PhD Student, Math Dept
Dr Simona Paoli	Australian Postdoctoral Fellow, Math Dept (from July 2005)
Dr Thorsten Palm	Scott Russell Johnson Memorial Fellow, Math Dept
Mr Craig Pastro	PhD Student, Math Dept
Mr Daniel Steffen	PhD Student, Math Dept
Dr Frank Valckenborgh	Macquarie University Research Fellow, Math Dept

Visitors since January 2004

Professor Richard Wood	Dalhousie University (Dec 2004 – Feb 2005)
Professor Aurelio Carboni	Università degli Studi dell'Insubria, Como, Italy (Jan 2005)
Dr Bertrand Toen	Laboratoire J.A. Dieudonne, University of Nice (April 2005)
Professor Robin Cockett	University of Calgary (April – Aug 2005)
Dr Vincent Schmitt	University of Leeds (25 March – 6 April 2005)
Ms Marni Sheppeard	University of Canterbury, New Zealand (Dec 2004)
Dr William Joyce	University of Canterbury, New Zealand (June 2004)
Professor George Janelidze	Mathematical Institute of the Academy of Science, Tbilisi, Georgia (Nov 2003 – April 2004); now University of Cape Town, South Africa
Mr Zurab Janelidze	State University of Georgia (Nov 2003 – Apr 2004)

Twelve Selected Highlights

1. Scott Russell Johnson Fellow **Michael Batanin** awarded an ARC Discovery Grant
2. A second Scott Russell Johnson Fellow **Thorsten Palm** appointed
3. A second generous donation made by Morgan–Phoa
4. Macquarie University Research Fellowship awarded to **Frank Valckenborgh**
5. Australian Postdoctoral Fellowship awarded to **Simona Paoli**
6. Minneapolis Workshop *n-Categories: Foundations and Applications* (7-18 June 2004)
7. **Dominic Verity**'s paper on *Complicial sets* finally completed and submitted
8. Organizers **Michael Batanin, Alexei Davydov, Mike Johnson, Steve Lack, and Amnon Neeman** announced the [StreetFest](#) to be held in July 2005
9. PhD candidate **Craig Pastro** arrived
10. A paper to form part of the PhD thesis of **Elango Panchadcharam** submitted
11. Street lectured on *Topos Theory* at the University of Canterbury (Christchurch, NZ)
12. [Quantum Group Graduate Course](#) TeX-ing completed by Vacation Scholar **Simon Byrne**

Stop Press

Saunders Mac Lane, who (jointly with Samuel Eilenberg) founded Category Theory 60 years ago, died on 14 April 2005 aged 95.

II. Short account of each researcher's contribution

The principal business of [CoACT](#) is research based in the area of mathematics called Category Theory.

1. [Ross Street](#) oversees all operations of CoACT: administration, finance and research.
2. [Michael Johnson](#) supports all operations, particularly financial. His research emphasizes applications to computer science.
3. [Max Kelly](#), as patriarch of Australian category theory, researches actively in enriched category theory and 2-dimensional universal algebra, and strongly supports the Australian Category Seminar ([ACS](#)).
4. [Wesley Phoa](#) is a highly valued industry link with a Cambridge University PhD in category theory.
5. [Dominic Verity](#) provides advice in accord with his experience in the finance industry and contributes deeply to research in higher category theory.
6. [Michael Batanin](#) is a star researcher in all aspects of CoACT's research and is strongly motivated by his experience in homotopy theory research.
7. [John Corbett](#) is primarily a mathematical physicist using category theory (particularly topos theory) as a foundational tool.
8. [Alexei Davydov](#) is a brilliant algebraist who develops the theory of monoidal (bi)categories motivated by his knowledge of group representation theory. With Steve Lack, he has taken over the organization of the [ACS](#).
9. [Brian Day](#) is an experienced mathematician actively researching enriched category theory — a subject he has deeply influenced since its inception — with inspiration from topological algebra.
10. [Carolyn Kennett](#) supports the [ACS](#) and her categorical research stems from the theory of simplicial sets.
11. [Steve Lack](#) is a brilliant category theorist and is a Chief Investigator of one of our ARC Discovery projects. He is a long-time organizer of the [ACS](#) and maintains the invaluable Web page <http://www.maths.usyd.edu.au/u/AusCat/>.
12. [Thorsten Palm](#) is a highly original contributor to higher category theory and strongly supports the [ACS](#).

13. [Frank Valckenborgh](#) uses category theory in physics; he collaborates with John Corbett and supports the [ACS](#).
14. [Catherine Menon](#), [Margaret Mitchell](#) are PhD students of Michael Johnson.
15. [Elango Panchadcharam](#), [Craig Pastro](#) and [Daniel Steffen](#) are PhD students of Ross Street.

More detailed research contributions are made explicit in the later sections.

III. Summary of research projects undertaken

While each paper prepared for publication (listed in Section VI) is a project in its own right, these smaller modules represent progress towards the greater goals covered by the following grant and personal projects.

Title: *Category theory arising from geometry, algebra, computer science and physics*

Personnel: Ross Street (Chief Investigator), Max Kelly (CI), Michael Johnson (CI), Stephen Lack (CI), Brian Day, Michael Batanin, Thorsten Palm, George Janelidze, Isar Stubbe, Zurab Janelidze, John Corbett., Frank Valckenborgh, Daniel Steffen, Catherine Menon.

Summary: Category theory is a branch of mathematics concerned with transformation and composition. It provides an algebra of wide-spread applicability for the synthesis of systems and processes in fields as diverse as geometry, physics and computer science, and also in mathematics itself. Often it can be used to clarify and simplify the learning, teaching and development of mathematics. The aim of this project is to develop the general theory of categories and specifically to investigate aspects appropriate to algebra, physics and computer science.

Our research on the project in 2004 exceeded expectations. There were 13 definitive publications representing the primary permanent measure of our scientific achievement. Added to this the activity of our Australian Category Seminar can be gleaned from its web site <http://www.maths.usyd.edu.au:8000/u/stevel/auscat/>. In [2], Street documented a new overview of descent theory and cohomology with coefficients in a weak higher-dimensional category. He studied Frobenius algebras, made the link with quantum groupoids, and documented it in [4], his first paper in a Physics journal. In a paper for a special workshop in Minneapolis at the Institute of Mathematics and its Applications, his goal was to set the stage for future development of higher-dimensional and enriched categories. Craig Pastro began in August 2004 as a postgraduate student whose PhD topic "Quantum categories and linear logic" is part of this project. Johnson and Street were on the scientific committee for the conference CT04 (U. British Columbia, Canada) and Street is an editor for the Proceedings which is in preparation. In the conference publication [8] and the expanded version [20], Lack and his collaborator Sobocinski defined a variant of extensive categories which is already proving influential in the graph-rewriting community. His [10] is related to crossed products, but has also turned out to have unexpected connections with theoretical computer science, as explained by Walters at the international conference CT04 in Vancouver. Kelly, after spending the first half of 2003 working with Lack on [6], which is the first stage of a proposed joint investigation on Flexible Monads, spent the rest of that year in and out of hospital, finally undergoing major spinal surgery. After recuperation, he worked with the visiting professor George Janelidze of Cape Town on object-actions in a semi-abelian category, generalizing the notions of semi-direct product for groups and Lie algebras; this has led to two papers with Professor F. Borceux of Louvain as a third author, the first of which is [19] while the second is about to be submitted. Meanwhile, Kelly's notes on categories with a class of colimits have been typed in LaTeX by Vincent Schmitt of Leicester, who will join in the completion and publication of this work.

Our research was spurred by visitors: William Joyce, Marni Sheppeard, George Janelidze, and Zurab Janelidze (Nov 2003 - April 2004) from the State University of Georgia.

There will be a major international conference at Macquarie University in July 2005 entitled *Categories in Algebra, Geometry and Mathematical Physics*; see <http://streetfest.maths.mq.edu.au/>. Approximately 100 international participants are expected.

Title: *Invariants of higher-dimensional categories, with applications*

Personnel: Ross Street (CI), Alexei Davydov, John Baez, Simona Paoli.

Summary: Complex systems in mathematics are difficult to tell apart so one constructs simpler structures from them. These structures must be equal, isomorphic or equivalent when the original systems are equivalent; the word invariant is used for such constructions. Higher-dimensional categories are complex structures that are currently gaining a lot of attention from mathematicians, physicists and computer scientists because of developing applications in those fields. This project will establish and study invariants for higher-dimensional categories which will be tested by examining their viability for producing results in group theory and homotopy theory.

The project proceeded as planned for 2004 and achieved further goals. The funds were used primarily to employ Dr Alexei Davydov as a research fellow. Davydov constructed a Gerstenhaber bracket on the extension algebra of the unit object of a monoidal category. If the monoidal category is braided this bracket is trivial but there is another bracket of degree two. This work was reported in talks at the Universities of Strasbourg [69] and Bristol [70], but still needs to be written up definitively. One way to obtain invariants of categorical structures is to construct richer structures from the given one where invariants are more easily found; this is the point of publications [12] and [3]. A structural invariant (cohomology) of crossed algebras was developed in [18]. The higher-dimensional Frobenius algebras of [4] are described in terms of a higher invariant form. Dominic Verity was finally motivated to complete his paper [27] characterizing strict omega-categories as simplicial sets; this identification of higher categories with “combinatorial spaces” allows the use for omega-categories of the known invariants for spaces. Many technical details and general principles of our work on this project were revealed in the Schrödinger Institute talks [67].

Elango Panchadcharam, as Street’s PhD student, is studying Mackey functors: these provide invariants for monoidal bicategories constructed from finite groups.

Title: *Foundations of higher dimensional homological algebra*

Personnel: Michael Batanin (CI)

Summary: *Homotopical Mathematics* is a term introduced recently to designate a rapidly developing methodology. It is based on the substitution of set theoretical notions by homotopy theoretical notions in a large part of mathematics relevant to geometry and physics. This approach has already produced spectacular applications in algebraic geometry, topology and mathematical physics. Homological algebra lies at the heart of this approach, yet its further development and application require clear and consistent foundations. The intention of this project is to construct such foundations, using methods of Higher Category Theory. As an outcome, proof of important conjectures from both areas will arise naturally.

Title: *Higher categorical structures in homotopy theory and homological algebra*

Personnel: Simona Paoli (CI)

Summary: This research is aimed at advancing homotopy theory and homological algebra through the use of higher categorical structures. One goal is to interpret cohomology classes in algebraic categories such as the category of commutative algebras. Another goal is to compare the use of established and more recent higher categorical structures – cat^n -groups and weak n -

categories – as homotopy models. The proposed approach to the interpretation of cohomology classes is simpler than the existing simplicial methods. Weak n -categories are an emerging field, with applications in diverse areas of mathematics. A comparison between the use of weak n -categories and cat^n -groups as homotopy models is unknown, and is imperative in view of applications.

Title: *Categorical universal algebra and the foundations of information management*

Personnel: Michael Johnson (CI), Jason Rennie, Catherine Menon (in 2004).

Summary: Information management depends upon providing structures on data stored in databases, semi-structured documents, formal specifications and libraries. Different structured arrangements of equivalent data frequently lead to apparent incompatibilities and limit the interoperations possible between systems. Categorical universal algebra has confronted this problem of inequivalent specifications of equivalent data in the context of algebraic structures. In preliminary work we have shown how analogous categorical techniques for the mathematical specification and analysis of structured data can be used to obtain data invariants which aid in the development of system interoperations. This project develops those techniques and extends them to semi-structured data and formal specifications.

Title: *Operadic techniques in the theory of crossed modules and hypercrossed complexes*

Personnel: Stephen Lack (CI), Simona Paoli.

Summary: Geometry concerns the spatial relationships between objects in the real world. Algebra involves calculations and formal manipulations of symbols; it has been described as a Faustian offer made to the geometer, since it provides powerful tools, but at the risk of eroding geometric intuition. Crossed modules and hypercrossed complexes are examples of such tools. This project represents a strategy to cheat the devil, by developing these tools in a context in which the geometric intuition is retained.

Title: *Higher-dimensional categories involving dendrotopes*

Personnel: Thorsten Palm

Summary: Palm's research on definitions of higher-dimensional categories involving dendrotopes has led him to study the broader notion of polytope and polytopic set. He has compiled a list of elementary constructions and results that should be more widely known among category theorists working in the area. One theorem concerns the representability of categories of polytopic sets, and more generally categories admitting a 'slice-object construction', as full categories of set-valued functors. It particularly yields a new and much shorter proof of the fact that dendrotopic sets form a presheaf category. Currently he is trying to apply the theorem to a certain full category of infinity-computads. Defined similarly to that of dendrotopic sets, it is as wide as to comprise not only R. Street's parity complexes, M. Johnson's pasting schemes and R. Steiner's directed complexes, but also structures without global acyclicity, such as general simplicial and cubical sets (without degeneracies) and, importantly, dendrotopic sets. Papers are in progress.

Title: *Omega-algebra*

Personnel: William Joyce

Summary: This work [13], [14] lays the foundation for an extension of super Lie algebra which allows for the first time quantisation of fermionic colour. This introduces a generalisation of algebra. The universal envelope for recoupling algebra has this structure. A Poincare-Birkhoff-Witt theorem has been

proved and the Hopf algebra structure uncovered. Current work is to develop its representation theory and the general categorical structure called omega-monoidal category theory.

Title: *Sheaves on a non-commutative topology*

Personnel: Isar Stubbe

Summary: This work could also be called 'logic having truth values in a quantaloid'. Applications in non-commutative algebra, quantum physics and quantum computing are being developed. The work of R.F.C. Walters and R. Street first hinted at the theory of such sheaves on a non-commutative topology as a theory of certain quantaloid-enriched categories; Stubbe's work may well be considered to be a continuation thereof. By investigating common mathematical foundations for non-commutative algebra, quantum physics and quantum computing, he hopes to gain a better insight to the very specific flavour of the logic (sometimes called "dynamic logic") that reigns those fields. He has in mind applications to the theory of quantum computing.

IV. Existing and potential internal and external linkages and collaborative arrangements

Section I makes many linkages apparent by noting the members, their departments, their institutions, and their visitors. CoACT has significant linkages with researchers in Montréal, Halifax, Milano, Chicago, Riverside, Cambridge (U.K.), Louvain-la-neuve, and San Jose (including Stanford University and some companies). For example, our visitor Isar Stubbe commented that "Collaboration between Louvain-la-neuve and Sydney [category theorists] goes back a long way. Notably, for many young Lovanists like myself the trip to Sydney has had decisive consequences for both the form and content of their PhD theses."

A linkage between CoACT and Professor Amnon Neeman's group at the Australian National University strengthened during 2004. Neeman was elected to Fellowship of the Australian Academy of Science in the most recent round.

V. Progress in relation to agreed performance indicators (A–G below) specified in the Centre's strategic plan

See Appendix 2.

VI. List of Centre publications, materials submitted for publication, provisional patents and other forms of commercialisation, and other measures of research output, including evidence of impact (e.g. citations, uptake of research developments by other groups, media reports, etc).

1. The long-term contribution of CoACT to the field was clearly recognized during the Workshop on "n-Categories: Foundations and Applications" from 7 June to 18 June 2004 at the Institute for Mathematics and its Applications (IMA at University of Minnesota, Minneapolis, MN 55455).
2. The international response to the [StreetFest](#) planned for July 2005 has been very pleasing.
3. The importance of Category Theory has been reflected in the numerous obituaries for Saunders Mac Lane in major international newspapers.
4. CoACT was further recognized by the inclusion of its Director in the *International Advisory Committee on Category Theory Conferences*.

Published since early 2004

- [1] Brian Day and Ross Street, Quantum categories, star autonomy, and quantum groupoids, in "Galois Theory, Hopf Algebras, and Semiabelian Categories", *Fields Institute Communications* **43** (American Math. Soc. 2004) 187–226.
- [2] Ross Street, Categorical and combinatorial aspects of descent theory, *Applied Categorical Structures* **12** (2004) 537–576.
- [3] Ross Street, The monoidal centre as a limit, *Theory and Applications of Categories* **13** (ed. G. Janelidze, S. Lack, F.W. Lawvere, E.M. Vitale and R.J. Wood: volume dedicated to Aurelio Carboni for his 60th birthday; 2004) 184–190.
- [4] Ross Street, Frobenius monads and pseudomonoids, *J. Math. Physics* **45(10)** (October 2004) 3930–3948.
- [5] Ross Street, Cauchy characterization of enriched categories, *Reprints in Theory and Applications of Categories* **4** (2004) 1–16.
- [6] G.M. Kelly and Stephen Lack, Monoidal functors generated by adjunctions, with applications to transport of structure, *Fields Inst. Commun.*, **43** (2004), 319–340.
- [7] Stephen Lack, The 3-by-3 lemma for regular Goursat categories, *Homology, Homotopy and Applications* **6**(2004) 1–3.
- [8] Stephen Lack and Pawel Sobocinski, Adhesive categories, in Foundations of Software Science and Computation Structures: 7th International Conference, FOSSACS 2004, *Lecture Notes in Computer Science* **2987** (Springer-Verlag, 2004) 273–288.
- [9] Stefano Kasangian, Stephen Lack, and Enrico M. Vitale, Coalgebras, braidings, and distributive laws, *Theory and Applications of Categories* **13**(2004) 129–146.
- [10] Stephen Lack, Composing PROPs, *Theory and Applications of Categories* **13** (2004) 147–163.
- [11] Stephen Lack, A Quillen model structure for bicategories, *K-theory* **33** (2004) 185–197.
- [12] Alexei Davydov, Braids and linear algebra, *J. Algebra* **278(2)** (2004) 725–738.
- [13] W. P. Joyce, Recoupling Lie algebra and universal omega-algebra, *J. Math. Phys.* **45(10)** (October 2004) 3859–3877.
- [14] W.P. Joyce, The next step beyond graded Lie algebra: recoupling Lie algebra, *Czechoslovak J. Phys.* **54(11)** (2004) 1335–1339.
- [15] W.P. Joyce, Braided premonoidal Mac Lane coherence, *J. Pure Appl. Algebra* **190** (2004) 155–176.
- [16] Isar Stubbe, Categorical structures enriched in a quantaloid: categories, distributors, functors, *Theory and Applications of Categories* **14** (2005) 1–45.

Accepted for publication

- [17] M. Batanin, The Eckmann-Hilton argument, higher operads and E_n -spaces, *Advances in Math.* (to appear).
- [18] A. Davydov, Cohomology of crossed algebras, *Contemporary Mathematics* (to appear).
- [19] F. Borceux, G. Janelidze and G.M. Kelly, Internal object actions, *Comment. Math. Univ. Carolinae* **46** (to appear 2005).
- [20] Stephen Lack and Pawel Sobocinski, Adhesive and quasiadhesive categories, *Theoretical Informatics and Applications* (to appear).

- [21] Stephen Lack and Simona Paoli, An operadic approach to internal structures, *Applied Categorical Structures* (to appear).
- [22] Stephen Lack, Limits for lax morphisms, *Applied Categorical Structures* (to appear).
- [23] Ross Street, An Australian conspectus of higher categories (to appear in the Proceedings of the [2004 Summer Program: n-Categories: Foundations and Applications, 1-18 June 2004](#) at the IMA of the University of Minnesota, Minneapolis).
- [24] Isar Stubbe, Categorical structures enriched in a quantaloid: regular presheaves, regular semicategories, *Cahiers de Topologie et Géométrie Différentielle Catégoriques*.
- [25] Isar Stubbe, Categorical structures enriched in a quantaloid: orders and ideals over a base quantaloid, *Applied Categorical Structures*.
- [26] Isar Stubbe, Towards 'dynamic domains': totally continuous cocomplete Q-categories, *Electronic Notes in Theoretical Computer Science* (Proceedings of MFPS XXI).

Submitted

- [27] Dominic Verity, Complicial sets, <http://arxiv.org/pdf/math.CT/0410412>
- [28] F. Borceux, G. Janelidze and G.M. Kelly, On the representability of actions in a semi-abelian category, *Theory and Applications of Categories* (submitted).
- [29] J.R.B. Cockett and Stephen Lack, Restriction categories III: colimits, partial limits, and extensivity (submitted).
- [30] Brian Day, Elango Panchadcharam and Ross Street, Lax braidings and the lax centre (submitted).
- [31] M. Batanin, Computads and slices of operads, <http://arxiv.org/pdf/math.CT/0209035>; submitted to *Theory and Appl. of Categories*.
- [32] M. Batanin, The combinatorics of iterated loop spaces, <http://arxiv.org/pdf/math.CT/0301221> ; submitted to *Topology and its Appl.*
- [33] Ross Street, *Quantum Groups: an entrée to modern algebra*, Notes from a Graduate Lecture Course (Macquarie University, March-June 1990); submitted as a book.
- [34] Isar Stubbe, The canonical topology on a meet-semilattice, submitted to the *International Journal of Theoretical Physics*.

In Preparation

- [35] R. Buchweitz, A. Davydov and R. Street, The Gerstenhaber homotopy in a monoidal bicategory (in preparation).
- [36] Ross Street, Monoidal actions, enriched categories, and convolution (in preparation).
- [37] Isar Stubbe and B. Van Steirteghem, Hilbert lattices; for the *Handbook of Quantum Logic* (Elsevier).
- [38] G.M. Kelly, S. Lack and A.J. Power, Flexibility for 2-monads.
- [39] F. Borceux, G. Janelidze, and G.M. Kelly, On the representability of actions.
- [40] A. Carboni, G. Janelidze, G.M. Kelly and Stephen Lack, Pointing a category .
- [41] G. Janelidze, G.M. Kelly and Stephen Lack, Factorization systems and Galois theory in the 2-dimensional context.
- [42] B.J. Day and Stephen Lack, Limits of small functors.
- [43] Michael Batanin and Mark Weber, Multitensors and higher dimensional operads.
- [44] Michael Batanin, Coherence, cooperative games and shuffle polytopes.
- [45] Michael Batanin, Brian Day and Ross Street, Lax globular monoidal functors out of Ω .

- [46] Michael Batanin, Clemens Berger, and Sjoerd Crans, Contractibility of the operad for Crans' 4-categories.
- [47] Alexei Davydov, Nuclei for pseudo-monoidal categories.
- [48] Alexei Davydov, Gerstenhaber structures on extensions.
- [49] Alexei Davydov, A_∞ -structures and Hochschild cohomology.
- [50] Michael Johnson, Rewriting techniques and coherence theorems.
- [51] Michael Johnson, David Naumann and John Power, Category Theoretic Models of Data Refinement.
- [52] Michael Johnson and Robert Rosebrugh, Universal view updatibility.
- [53] G.M. Kelly and M.-C. Pedicchio, On one-sortedness of algebraic categories.
- [54] G.M. Kelly and A.J. Power, Enrichment for monads on the category of categories.
- [55] F. Borceux and G.M. Kelly, On accessibility for enriched categories.
- [56] B.J. Day and G.M. Kelly, On categories with a distributive law.
- [57] G.M. Kelly and B. Mesablishvili, On enriched monads of descent type and of effective descent type.
- [58] Alexei Davydov, Quasi-commutative monoids.
- [59] Alexei Davydov, Generators and relations for categories of representations of symmetric groups.

Conference presentations and lecture series

- [60] S. Lack, Quillen model structures for 2-categories and bicategories, International Workshop "n-categories: foundations and applications" at the Institute for Mathematics and its Applications, Minneapolis (June 2004).
- [61] S. Lack, Composing PROPs, at CT04, University of British Columbia (July 2004).
- [62] S. Lack, Pure-braided monoidal categories, Annual Meeting of the AustMS, RMIT (September 2004).
- [63] R. Street, An Australian conspectus of higher categories, International Workshop "n-categories: foundations and applications" at the Institute for Mathematics and its Applications, Minneapolis (June 2004).
- [64] R. Street, Frobenius algebras and monoidal categories, Annual Meeting of the AustMS, RMIT (September 2004).
- [65] R. Street, Lectures on Topos Theory (eleven hours), Univ. Canterbury, Christchurch, NZ (November 2004).
- [66] M. Johnson, Partial morphisms, fibrations and database specifications, Foundational Methods in Computer Science (FMCS04) at University of Calgary (June 2004).
- [67] A. Davydov, K-theory of braided monoidal categories, ESI program "Tensor categories in Mathematics and Physics" Ervin Schrödinger Institute, Vienna, Austria (June-July 2004).
- [68] A. Davydov, Quasi-commutative algebras, "Non-commutative Geometry and Representation Theory in Mathematical Physics", Karlstad University, Sweden (July 2004).
- [69] A. Davydov, Gerstenhaber structures on extensions, IRMA, University of Strasbourg, France (November 2004).
- [70] A. Davydov, Gerstenhaber structures on extensions, University of Bristol, UK (December 2004).

VI. A financial summary for the year, including details of external grants and contracts, and projected cash flow and budget for the following year.

See Appendix 3.

VIII. The status of implementation of the recommendations in the latest review of the Centre.

As yet there has been no review.

IX. Proposed activities for the coming year, and related performance indicators.

- A. The [ACS](#) will continue. [StreetFest](#) can be considered a concentrated version of [ACS](#): the conference, in honour of Street's 60th birthday, will be held at Macquarie University 11–16 July 2005 (*Categories in Algebra Geometry and Mathematical Physics*) with a continuing satellite conference at the Australian National University 18–21 July 2005 (*Workshop on Categorical Methods in Algebra, Geometry and Mathematical Physics*). Approximately 100 international mathematicians will participate.
- B. Our 10 accepted and 8 submitted papers already exceed this benchmark for 2005.
- C. There will be a Proceedings volume arising from the [StreetFest](#) probably published by the American Math. Society. We will all continue on our editorial boards.
- D. Street will be an invited speaker for the 100th anniversary of the birth of Charles Ehresmann (October 2005, Amiens, France). Preceding this he will lecture at Louvain-la-neuve in Belgium (September 2005).
- E. International students are consistently approaching our members as PhD supervisors. Street is currently corresponding with 4 strong potential students. Simon Byrne is currently writing his Math. Honours project with Street.
- F. Street has applied for an ARC Discovery Grant for 2006; the project on *String Theory* proposes to employ Davydov
- G. Reviewing is expected to continue at the same rate. In particular, Street will review Johnstone's books "Sketches of an elephant: A topos theory compendium Volumes 1 and 2" for *Zentralblatt*.

Appendix 1: Detail on Section V Item D

Papers reviewed by Ross Street for Math. Reviews and Zentralblatt für Math.

13. MR2086080 Marmolejo, F. Distributive laws for pseudomonads. II. *J. Pure Appl. Algebra* 194 (2004), no. 1-2, 169–182.
14. MR2075598 (2005f:18005) Palm, Thorsten Dendrotopic sets. Galois theory, Hopf algebras, and semiabelian categories, 411–461, *Fields Inst. Commun.*, 43, Amer. Math. Soc., Providence, RI, 2004.
15. MR2075597 (2005d:18001) Martins-Ferreira, N. Weak categories in additive 2-categories with kernels. Galois theory, Hopf algebras, and semiabelian categories, 387–410, *Fields Inst. Commun.*, 43, Amer. Math. Soc., Providence, RI, 2004.
16. MR2071328 (2005d:18007) Carrasco, Pilar; Martínez-Moreno, Juan Simplicial cohomology with coefficients in symmetric categorical groups. *Appl. Categ. Structures* 12 (2004), no. 3, 257–285.
17. MR2059184 (2005a:18012) Forcey, Stefan Enrichment over iterated monoidal categories. *Algebr. Geom. Topol.* 4 (2004) 95–119.
18. MR2032029 (2005a:18011) Elgueta, Josep Cohomology and deformation theory of monoidal 2-categories. I. *Adv. Math.* 182 (2004), no. 2, 204–277.
19. MR2025593 (2005a:18008) Cheng, Eugenia Weak n -categories: opetopic and multitopic foundations. *J. Pure Appl. Algebra* 186 (2004), no. 2, 109–137.
20. MR2025588 (2005a:18009) Cheng, Eugenia Weak n -categories: comparing opetopic foundations. *J. Pure Appl. Algebra* 186 (2004), no. 3, 219–231.
21. Zbl 1055.18001 Dawson, R.J.M.; Pare, R.; Pronk, D.A., Free extensions of double categories. *Cah. Topol. Géom. Différ. Catég.* 45, No.1, 35–80 (2004).
22. Zbl 1055.18002 Marmolejo, F., Distributive laws for pseudomonads. II. *J. Pure Appl. Algebra* 194, No.1-2, 169–182 (2004).
23. Zbl 1052.18003 Joyce, William P. Braided premonoidal Mac Lane coherence. *J. Pure Appl. Algebra* 190, No.1-3, 155–176 (2004).
24. Zbl 1052.18004 Müger, Michael Galois extensions of braided tensor categories and braided crossed G -categories. *J. Algebra* 277, No.1, 256–281 (2004).
25. Zbl 1043.18004 MacDonald, John; Sobral, Manuela Aspects of monads. Pedicchio, Maria Cristina (ed.) et al., *Categorical foundations. Special topics in order, topology, algebra, and sheaf theory*. Cambridge: Cambridge University Press. *Encycl. Math. Appl.* 97, 213–268 (2004).
26. Zbl 1042.18004 Cheng, Eugenia Weak n -categories: Comparing opetopic foundations. *J. Pure Appl. Algebra* 186, No.3, 219–231 (2004).
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Appendix 2: Detail on Section V

Progress in relation to agreed performance indicators (A–G below) specified in the Centre's strategic plan

A. Activity of the Australian Category Seminar ([ACS](#))

Benchmark At least 40 three-hour seminars per annum.

28 three-hour Seminars were held in 2004. For details of dates, speakers, titles and some abstracts, see www.maths.usyd.edu.au:8000/u/stevel/auscat/titles-2004.html. In a sense, the Minneapolis Workshop could be viewed as a winter extension of [ACS](#).

B. Scholarly publication in international refereed journals

Benchmark Fifteen journal articles per annum.

16 publications have appeared in refereed journals since the end of 2003. Another 10 have been accepted. See Section VI for details.

C. Editing of papers submitted to scholarly journals and to special issue volumes

Benchmark One special issue volume per annum.

Lack and Janelidze were Editors for the volume of *Theory and Applications of Categories* dedicated to Professor Aurelio Carboni on his 60th birthday.

Street is an editor (with John MacDonald, George Janelidze and Walter Tholen) of the Proceedings of CT04 to appear in *Theory and Applications of Categories*.

Batanin and Lack were appointed to the Editorial Board of *Applied Categorical Structures* and Lack continues on the Board of *Theory and Applications of Categories*.

Street continued on the Editorial Boards of six international journals:

Advances in Mathematics; *Applied Categorical Structures*; *Theory and Applications of Categories*; *Homology, Homotopy and Applications*; *Bulletin of the Australian Mathematical Society*; *Cahiers de topologie et géométrie différentielle catégoriques*

Kelly continued on the Editorial Boards of: *Applied Categorical Structures* and *Theory and Applications of Categories*.

D. International plenary addresses and membership of conference scientific steering committees

Benchmark One international plenary address or one conference steering committee per annum.

1) A historically significant Workshop for CoACT's research was held as an Institute of Mathematics and its Applications Summer Program (University of Minnesota, Minneapolis, USA): the title was [n-Categories: Foundations and Applications](#) (7-18 June 2004) organized by Professors Peter May (U Chicago) and John Baez (U Calif. Riverside). The Australian contribution to this conference was major. Batanin, Davydov, Johnson, Lack and Street were all invited speakers and prime participants. Many expatriate Australians and CoACT visitors also provided fundamental input. Some plenary talks were:

Batanin: *The A_∞ operad definition of n -category*

Batanin: *Perspectives of operadic approaches*

Lack: *Higher model categories*

Street: *An Australian conspectus of higher categories*

2) Street was on the Scientific Committee for the annual category meeting CT04 in Vancouver (July 2004).

3) Steve Lack and Michael Batanin were invited speakers at the Conference CT04 in Vancouver.

4) Street has agreed to be on the Scientific Committee of the Conference: *The Mathematics of String Theory* at the ANU in 2007.

5) Street was invited by the Mathematics, Physics and Philosophy Departments of the University of Canterbury (Christchurch, New Zealand) to lecture on *Topos Theory*. He lectured for 11 hours over two days to an enthusiastic audience.

6) Street was appointed to a new *International Advisory Committee on Category Theory Conferences*; the full committee is: Bill Lawvere (Chair), Peter Johnstone (Vice-Chair), Jiri Adamek, Francis Borceux, Ross Street, Walter Tholen, Richard Wood

7) Batanin, Lack and Street all gave presentations at the Australian Math. Soc. Annual Meeting at RMIT in Melbourne.

E. Effective Full-Time Student Units (EFTSU) attracted and completion rates, especially for postgraduate students

Benchmark Ten postgraduate EFTSU and two PhD completions per annum.

Margaret Mitchell and Daniel Steffen are currently enrolled for PhD in their final year while Catherine Menon, Elango Panchadcharam, and Craig Pastro are also PhD students.

F. Funding attracted through grant applications and other sources

Benchmark \$300000 in external funding per annum.

See VII below: 81% of the benchmark. However, this figure excludes generous travel and living expense contributions from host institutions, most notably in 2004 from the IMA (Minneapolis). The outlook for 2005 is well over the benchmark even without special support from the Australian National University, the Australian Mathematical Society (AMSI) and the Macquarie University Mathematics Department.

G. Visibility in the scientific review literature

Benchmark 20 reviews written by CoACT members appearing in Mathematical Reviews, Zentralblatt für Mathematik, or other influential mathematical and computing reviewing journals.

Benchmark exceeded. At least 24 articles were reviewed for Math. Reviews and Zentralblatt; see Appendix 1 for more detail.